A Runtime Ethical Enforcement Framework for Autonomous Systems

SLEEC Evaluation with Experts

INTRODUCTION

Please note that this interview will be recorded for evaluation purposes only

Case Study

The ARI robot (https://pal-robotics.com/robot/ari) is deployed in a rehabilitation center to support patients diagnosed with diabetes who are undergoing rehabilitation and need to follow a strict diet and physical training program.

The main **mission** of ARI is to assist patients in adhering to their prescribed routines by providing personalized guidance, monitoring health-related data, and facilitating communication with healthcare staff when needed.

Specifically, this includes providing instructions on *which exercises* to perform and *the number of repetitions*, ensuring *timely delivery of meals according to each patient's dietary plan*, and supporting overall *adherence to medical recommendations*.

Case Study

Robot's tasks and capabilities:

- *Continuous Data Monitoring via Smartwatch*: each patient is equipped with a smartwatch that monitors key health metrics (e.g., heart rate, activity level, glucose trend). The smartwatch communicates exclusively with ARI, i.e., no other device or system has access to this data.
- *Exercise Support*: during physical training sessions, ARI uses its camera and motion recognition capability to monitor and count repetitions of the patient's exercises. It can provide verbal encouragement (through text-to-speech) and visual demonstrations on its tablet display.
- *Interaction with Patients*: ARI can understand spoken commands from patients via voice recognition.
- *Healthcare Communication Support*: if ARI detects an unexpected situation, it can call an external person, such as a nurse or physician, to provide immediate assistance.
- *Dietary Guidance*: at mealtimes, ARI can provide reminders (through text-to-speech) of the patient's dietary plan and help ensure they adhere to their prescribed diet and eat according to the scheduled times.
- *Patient Feedback*: during training sessions and mealtimes, ARI can receive a patient's feedback or preferences on its tablet display.

SLEEC Principles and Description

Privacy	Limiting intrusion on the personal space of the user and ensuring privacy is protected; safeguarding health data, practising good data stewardship, and granting or restricting access to medical records
Respect for Autonomy	Granting and withdrawing of permissions, including consent and assent; ensuring the user maintains an appropriate level of control
Dignity	Understanding and accommodating the user's social and cultural sensitivities, respectful treatment
Explainability and transparency	Informing the user about system decision-making and any inferences made; providing justification for a course of action adopted
Beneficence	Maximising good outcomes
Non-maleficence	Minimising harm by ensuring safety and reducing the possibility of physical and psychological harm to the user

Townsend, B., et al. "From pluralistic normative principles to autonomous-agent rules." Minds and Machines 32.4 (2022): 683-715.

DISCUSSION

Q1. **Is the SLEEC rule correct and effective in preventing or reducing system's unethical behavior?**

Evaluating Understandability, Correctness and Completeness

*for each SLEEC rule

Q2. **Are the principles correctly mapped to the SLEEC rule, the base rule, and the hedge clauses? Are there any principles that are not covered by this mapping?**

Evaluating Correctness

*for each SLEEC rule

Q3. **Is the proposed rule correctly labeled as Social, Legal, Ethical, Empathetic, and Cultural?**

Evaluating Correctness

*for each SLEEC rule

Q4. **Are there any SLEEC rules you consider missing?**

Evaluating Completeness

Anything to add?

THANKS!